

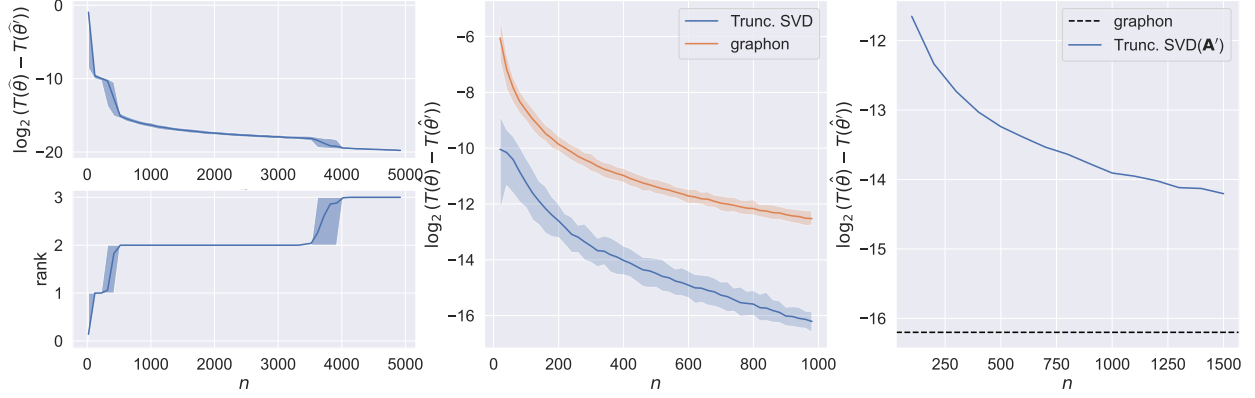


Figure 1: *Upper Left*: difference between target functions for optimal interventions of a network on n vertices sampled from $(1, 1/2)$ -Hölder graphon $W_1(x, y) = \sqrt{|x - y|}$ and interventions based on hard-thresholding estimator. *Lower Left*: Rank of the hard-thresholding estimator for a network on n vertices sampled from $W_1(x, y)$. *Middle*: Difference between target functions for optimal interventions of a network on n -vertices sampled from SBM with 4 communities and interventions computed using a) graphon b) hard-thresholding estimator. *Right*: The difference between target functions for optimal interventions of a network $\mathbf{A}$ of size 10000 sampled from SBM with 4 communities and interventions based on a) graphon b) hard-thresholding estimator computed for other network $\mathbf{A}'$ of size n .

6 Conclusion

Driven by applications in graphon games, we developed an estimator for an unknown graphon based on a sampled network. Under standard regularity conditions for the graphon, our estimator possesses two key features: (i) it approximates the unknown graphon in the operator norm, and (ii) it is of low rank. We established upper bounds for the convergence rate and rank of the estimator. Furthermore, we demonstrated that the estimator yields near-optimal solutions for the social welfare problem in linear-quadratic graphon games, with efficient computation. We also quantified the convergence rates of the associated costs and the computational complexity relative to the network size.

References

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